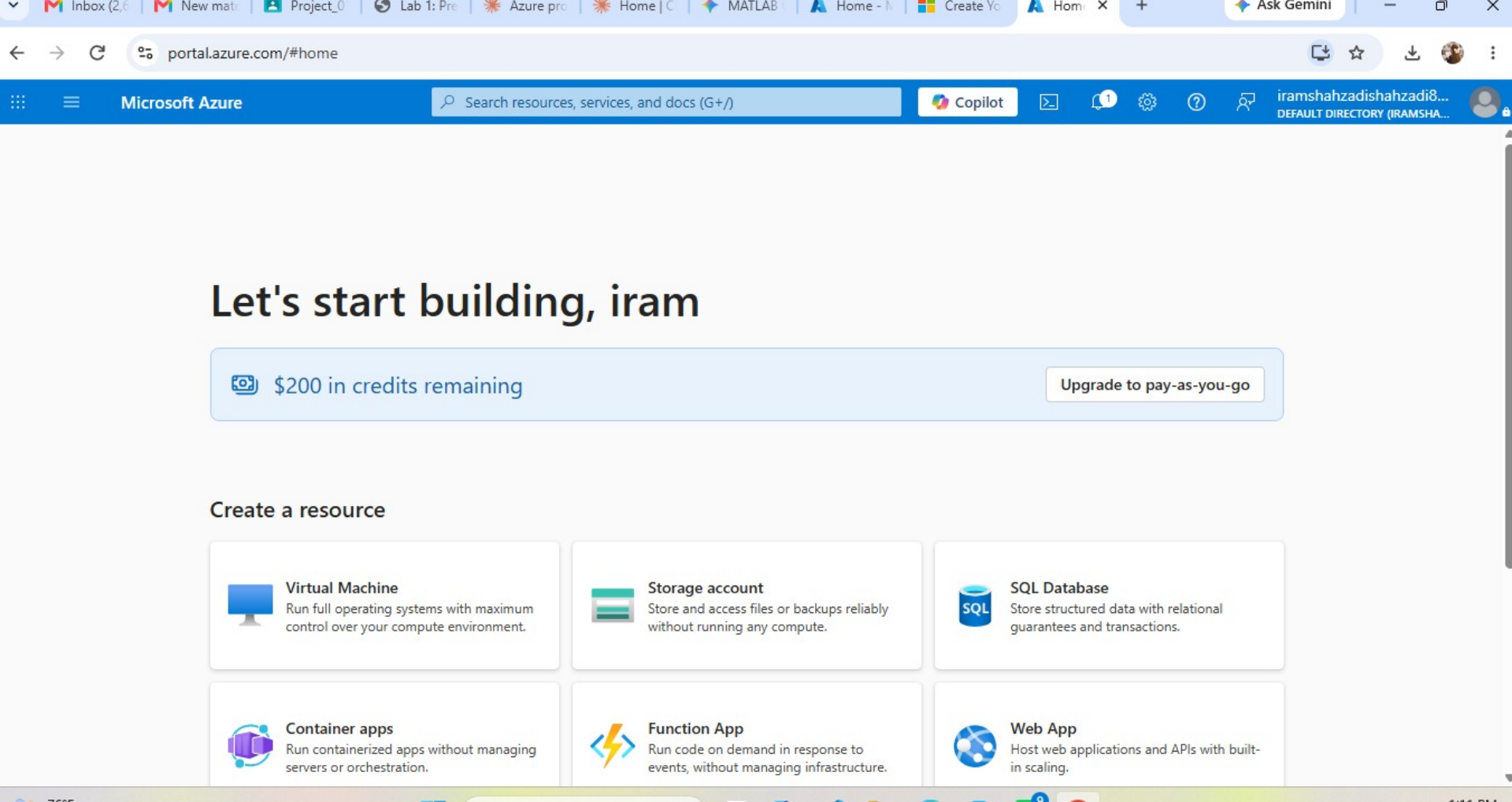




# Let's start building, iram



\$200 in credits remaining

Upgrade to pay-as-you-go

## Create a resource



### Virtual Machine

Run full operating systems with maximum control over your compute environment.



### Storage account

Store and access files or backups reliably without running any compute.



### SQL Database

Store structured data with relational guarantees and transactions.



### Container apps

Run containerized apps without managing servers or orchestration.



### Function App

Run code on demand in response to events, without managing infrastructure.



### Web App

Host web applications and APIs with built-in scaling.

Home &gt; Resource Manager

# Resource Manager | Resource groups

Default Directory



Summarize my costs by service

How to manage changes with deployment tools?

Export resource groups using Bicep or Terraform



Search

Resource Manager

All resources

Favorite resources

Recent resources

Resource groups

Tags

&gt; Organization

&gt; Tools

&gt; Deployments

&gt; Help



Create



Manage view



Refresh



Export to CSV



Open query



Assign tags



Add to service group



Group by none



Filter for any field...

Subscription equals all

Location equals all



Add filter



Name ↑

Subscription

Location

 rg-mlab1-iram15

Azure subscription 1

East US

[Home](#) > [Azure Machine Learning](#)

# Azure Machine Learning

Create a machine learning workspace



Validation passed

[Basics](#)   [Inbound Access](#)   [Outbound Access](#)   [Encryption](#)   [Identity](#)   [Tags](#)   [Review + create](#)

|                      |                                        |
|----------------------|----------------------------------------|
| Subscription         | Azure subscription 1                   |
| Resource group       | rg-mlab1-iram15                        |
| Region               | East US                                |
| Name                 | mlw-lab1-iram15                        |
| Storage account      | (new) mlwlab1iram152606256493          |
| Key vault            | (new) mlwlab1iram151030151340          |
| Application insights | (new) mlwlab1iram155230222237          |
| Container registry   | None                                   |
| Connectivity method  | Enable public access from all networks |
| Network isolation    | Public                                 |

portal.azure.com/#@iramshahzadishahzadi892gmai.onmicrosoft.com/resource/subscriptions/f3366df5-0550-438c-b31e-e437667ba4b0/resourcegroups/rg-mlab1...

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

iramshahzadishahzadi8...  
DEFAULT DIRECTORY (IRAMSHA...

Home > Microsoft.MachineLearningServices | Overview

mlw-lab1-iram15  
Azure Machine Learning workspace

Search

Download config.json

Delete

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Events

Settings

Monitoring

Automation

Support + troubleshooting

Essentials

Resource group : rg-mlab1-iram15

Location : East US

Subscription : Azure subscription 1

Storage : mlwlab1iram155001683223

Provisioning State : Succeeded

Studio web URL : https://ml.azure.com?tid=49846e27-c863-482f-abaa-3a8090395231&wsid=/...

Container Registry : ...

Key Vault : mlwlab1iram152708731420

Application Insights : mlwlab1iram159628243326

MLflow tracking URI : azureml://eastus.api.azureml.ms/mlflow/v1.0/subscriptions/f3366df5-0550-43...

JSON View

Work with your models in Azure Machine Learning Studio

The Azure Machine Learning Studio is a web app where you can build, train, test, and deploy ML models. Launch it now to start exploring, or learn more about the Azure Machine Learning studio

Launch studio

- All workspaces
- Home
- Model catalog
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# mlw-lab1-iram15

## Generative AI with Prompt flow

## Generative AI models

muse-glimmer-30b

Chat completion

gpt-live-1

Audio generation

rerank-2.5-lite

Text classification

DeepSeek-V4.1-Flash

Chat completion

## Notebook samples



### Get started: Train and deploy a model

Train and deploy a sample image classification model.

Start

25 minutes



### Index and search your own data with GPT

Bring your own data to look up using GPT with LangChain

Start

20 minutes



### Distributed GPU training

Run a sample multi-GPU image classification experiment.

Start

30 minutes

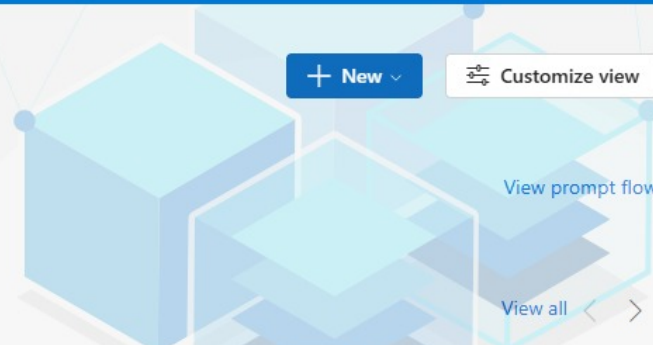


### Automate with Pipelines

Create a production pipeline for credit default prediction

Start

30 minutes



×

## 2 Scheduling

3 Security optional

#### 4 Applications optional

**S** **Tags**  
optional

## 6 Review

Select the name and virtual machine size you would like to use for your compute instance

① Note that a compute instance can not be shared. It can only be used by a single assigned user. By default, it will be assigned to the creator and you can change this to a different user in the Security step.

ci-iram15-lab1

☒ CPU   ☐ GPU

☒ Select from recommended options ☐ Select from all options

| Name ↑                                                                               | Category           | Workload types                                                   | Available quota ⓘ | Cost ⓘ    |
|--------------------------------------------------------------------------------------|--------------------|------------------------------------------------------------------|-------------------|-----------|
| <input checked="" type="radio"/> Standard_DS11_v2<br>2 cores, 14GB RAM, 28GB storage | Memory optimized   | Development on Notebooks (or other IDE) and light weight testing | 6 cores           | \$0.18/hr |
| <input type="radio"/> Standard_DS3_v2<br>4 cores, 14GB RAM, 28GB storage             | General purpose    | Classical ML model training on small datasets                    | 6 cores           | \$0.29/hr |
| <input type="radio"/> Standard_E4ds_v4<br>4 cores, 32GB RAM, 150GB storage           | Memory optimized   | Data manipulation and training on medium-sized datasets (1-10GB) | 4 cores           | \$0.29/hr |
| <input type="radio"/> Standard_F4s_v2<br>4 cores, 8GB RAM, 32GB storage              | Compute optimiz... | Data manipulation and training on large datasets (> 10 GB)       | 6 cores           | \$0.17/hr |

Cancel





## Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).

Required fields are marked with an asterisk (\*).

### 1 General

Owner \*



Repository name \*

azureml-knn-maintenance

✓ azureml-knn-maintenance is available.

Great repository names are short and memorable. How about [bookish-carnival](#)?

Description

KNN predictive maintenance on Azure ML

38 / 350 characters

### 2 Configuration

Choose visibility \*

Choose who can see and commit to this repository

 Public

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

On ☒

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

 Python

Add license

Licenses explain how others can use your code. [About licenses](#)

No license

Create repository

GitHub Apps

OAuth Apps

Personal access tokens

Fine-grained tokens

Tokens (classic)

## New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

### Note

azureml-lab1

What's this token for?

### Expiration

30 days (Oct 29, 2026)

The token will expire on the selected date

### Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

|                                                     |                                                                     |
|-----------------------------------------------------|---------------------------------------------------------------------|
| <input checked="" type="checkbox"/> repo            | Full control of private repositories                                |
| <input checked="" type="checkbox"/> repo:status     | Access commit status                                                |
| <input checked="" type="checkbox"/> repo_deployment | Access deployment status                                            |
| <input checked="" type="checkbox"/> public_repo     | Access public repositories                                          |
| <input checked="" type="checkbox"/> repo:invite     | Access repository invitations                                       |
| <input checked="" type="checkbox"/> security_events | Read and write security events                                      |
| <input type="checkbox"/> workflow                   | Update GitHub Action workflows                                      |
| <input type="checkbox"/> write:packages             | Upload packages to GitHub Package Registry                          |
| <input type="checkbox"/> read:packages              | Download packages from GitHub Package Registry                      |
| <input type="checkbox"/> delete:packages            | Delete packages from GitHub Package Registry                        |
| <input type="checkbox"/> admin:org                  | Full control of orgs and teams, read and write org projects         |
| <input type="checkbox"/> write:org                  | Read and write org and team membership, read and write org projects |



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Default Directory &gt; mlw-lab1-iram15 &gt; Notebooks

## Notebooks

Files

Samples



Users

iramshahzadishahzadi892

1:ci-iram15-lab1 x



Compute: ci-iram15-lab1 - Running

Welcome to the Azure Machine Learning terminal

Enter "git clone [url]" to clone a repo

Enter "git --help" to learn about Git CLI. To learn more about integrating Git with the Azure Machine Learning terminal, navigate here:  
<https://learn.microsoft.com/en-us/azure/machine-learning/concept-train-model-git-integration#clone-git-repositories-into-your-workspace-file-system>

Enter "az ml --help" to learn about Azure ML CLI v2

Note: Use "az login --identity" instead of "az login" to avoid device code authentication

(azureml\_py38) azureuser@ci-iram15-lab1:~/cloudfiles/code/Users/iramshahzadishahzadi892\$

← → ↺

ml.azure.com/fileexplorerAzNB?wsid=/subscriptions/f3366df5-0550-438c-b31e-e437667ba4b0/resourcegroups/rg-mlab1-iram15/providers/Microsoft.MachineLearning...

🔍 ☆

Microsoft Foundry | Azure Machine Learning

🕒 2 ⚙️ 10 ? 😊

Azure subscription 1  
mlw-lab1-iram15

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>... Prompt flow

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1:ci-iram15-lab1

☰ 🖥 ? 📄

● Compute: ci-iram15-lab1 - Running

🔍 + 📄

📄 Users

iramshahzadishahzadi892

```
Command 'create' not found, did you mean:
  command 'pcreate' from deb pbuilder-scripts (22)
Try: sudo apt install <deb name>
bash: syntax error near unexpected token `azureuser@ci-iram15-lab1:~/cloudfiles/code/Users/iramshahzadishahzadi892/azureml-knn-maintenance$'
Username: command not found
Command 'Password' not found, did you mean:
  command 'assword' from deb assword (0.12.2-1)
Try: sudo apt install <deb name>
remote:: command not found
fatal:: command not found
bash: syntax error near unexpected token `azureuser@ci-iram15-lab1:~/cloudfiles/code/Users/iramshahzadishahzadi892/azureml-knn-maintenance$ghp_rDSmYh7XBgwdOfw
9LWwAkp707eKzVA090cCH'
(azureml_py38) azureuser@ci-iram15-lab1:~/cloudfiles/code/Users/iramshahzadishahzadi892/azureml-knn-maintenance$ git push origin main
Username for 'https://github.com': Iram-Shahzadii
Password for 'https://Iram-Shahzadii@github.com':
(azureml_py38) azureuser@ci-iram15-lab1:~/cloudfiles/code/Users/iramshahzadishahzadi892/azureml-knn-maintenance$ git push origin main
Username for 'https://github.com': Iram-Shahzadii
Password for 'https://Iram-Shahzadii@github.com':
(azureml_py38) azureuser@ci-iram15-lab1:~/cloudfiles/code/Users/iramshahzadishahzadi892/azureml-knn-maintenance$ git push https://Iram-Shahzadii:ghp_zJOCk7K5k
kKHjHnnP8DLmOvOBZxNu20Fz8Vp@github.com/Iram-Shahzadii/azureml-knn-maintenance.git main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 659 bytes | 25.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Iram-Shahzadii/azureml-knn-maintenance.git
11688e5..9eef027  main -> main
(azureml_py38) azureuser@ci-iram15-lab1:~/cloudfiles/code/Users/iramshahzadishahzadi892/azureml-knn-maintenance$ ^C
(azureml_py38) azureuser@ci-iram15-lab1:~/cloudfiles/code/Users/iramshahzadishahzadi892/azureml-knn-maintenance$ git push https://Iram-Shahzadii:ghp_gwgij05Z7
rIwFHPceWpIe97mFyyjMr4aXBR0@github.com/Iram-Shahzadii/azureml-knn-maintenance.git mai^C
(azureml_py38) azureuser@ci-iram15-lab1:~/cloudfiles/code/Users/iramshahzadishahzadi892/azureml-knn-maintenance$ git push origin main
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/Iram-Shahzadii/azureml-knn-maintenance.git/'
(azureml_py38) azureuser@ci-iram15-lab1:~/cloudfiles/code/Users/iramshahzadishahzadi892/azureml-knn-maintenance$ git push origin main
Username for 'https://github.com': Iram-Shahzadii
Password for 'https://Iram-Shahzadii@github.com':
```

azureml-knn-maintenance Public

Pin Watch 0 Fork 0 Star 0

main 1 Branch 0 Tags

Go to file

Add file

Code

|                  |                                           |                          |           |
|------------------|-------------------------------------------|--------------------------|-----------|
| Iram-Shahzadii   | Lab 1: project structure and requirements | 9eef027 · 20 minutes ago | 2 Commits |
| automi           | Lab 1: project structure and requirements | 20 minutes ago           |           |
| data             | Lab 1: project structure and requirements | 20 minutes ago           |           |
| deployment       | Lab 1: project structure and requirements | 20 minutes ago           |           |
| designer         | Lab 1: project structure and requirements | 20 minutes ago           |           |
| notebooks        | Lab 1: project structure and requirements | 20 minutes ago           |           |
| screenshots      | Lab 1: project structure and requirements | 20 minutes ago           |           |
| .gitignore       | Lab 1: project structure and requirements | 20 minutes ago           |           |
| README.md        | Initial commit                            | 41 minutes ago           |           |
| requirements.txt | Lab 1: project structure and requirements | 20 minutes ago           |           |

README

# azureml-knn-maintenance

## About

KNN predictive maintenance on Azure ML

Readme

Activity

0 stars

0 watching

0 forks

## Releases

No releases published

[Create a new release](#)

## Packages

No packages published

[Publish your first package](#)

## Contributors 1

Iram-Shahzadii Iram Shahzadi

Microsoft Foundry | Azure Machine Learning

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00\_prepare\_data.ipyn

Edit in VS Code

Compute: ci-iram15-lab1 - Running

Python 3.10 - SDK v2

You need to be authenticated to the compute to use any Azure SDK. Please use the authenticate button to get authenticated.

Authenticate

ci-iram15-lab1 · Kernel launching

Last saved a few seconds ago

Python 3.10 - SDK V2

1

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1:ci-iram15-lab1 x 00\_prepare\_data.ipynb x

...

Edit in VS Code

Compute: ci-iram15-lab1 - Running

Python 3.10 - SDK v2

Successfully authenticated to the compute. You can now use any Azure SDK.

ci-iram15-lab1 · Kernel idle Last saved a few seconds ago Python 3.10 - SDK v2

... dtype: object

Target balance (share of rows):

0 0.9661

1 0.0339

Name: machine\_failure, dtype: float64

1 df.to\_csv("../data/ai4i\_clean.csv", index=False)

2 print("Saved", df.shape, "to data/ai4i\_clean.csv")

[S] ✓ <1 sec

Saved (10000, 7) to data/ai4i\_clean.csv

1

## Notebooks

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- Users
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    - azureml-knn-maintenance
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1:ci-iram15-lab1

00\_prepare\_data.ipynb



Edit in VS Code



Compute:

ci-iram15-lab1

Running



ci-iram15-lab1 · Kernel idle

Last saved a few seconds ago



```
1 from azure.ai.ml.entities import Data
2 from azure.ai.ml.constants import AssetTypes
3
4 table_asset = Data(
5     name="ai4i-maintenance-table",
6     version="1",
7     type=AssetTypes.MLTABLE,
8     path="../data/mltable_ai4i",
9     description="AI4I 2020 predictive maintenance as MLTable for AutoML",
10 )
11 table_asset = ml_client.data.create_or_update(table_asset)
12 print("Registered:", table_asset.name, "version", table_asset.version)
```

[12] ✓ 3 sec

... Uploading mltable\_ai4i (0.3 MBs): 100% [██████████] 295336/295336 [00:00&lt;00:00, 7800363.76it/s]

Registered: ai4i-maintenance-table version 1



1

## Data

Data assets

Datatstores

Data import

PREVIEW

Data connections

PREVIEW

Data assets are immutable references to your data that can be created from datatstores, local files, public URLs, or Open Datasets. Data assets created with AzureML v2 APIs cannot be deleted, but you can up-version or archive them for easy referencing and reuse in machine learning tasks. Deleting data assets created with v1 APIs will permanently delete the data asset and all metadata. [Learn more about data assets](#)

+ Create

Refresh

Archive

Reset view



Show latest version only



Include archived



View my data

Search

Filter

Columns

| Name                   | ☆ | Ve... | Data source        | Created on ↓         | Modified on          | Type  | Propert... | Created by    | Tags |
|------------------------|---|-------|--------------------|----------------------|----------------------|-------|------------|---------------|------|
| ai4i-maintenance-table |   | 1     | workspaceblobstore | Sep 29, 2026 3:32 PM | Sep 29, 2026 3:32 PM | Table |            | iram shahzadi |      |
| ai4i-maintenance-csv   |   | 1     | workspaceblobstore | Sep 29, 2026 3:20 PM | Sep 29, 2026 3:20 PM | File  |            | iram shahzadi |      |

```
type,air_temp_k,process_temp_k,rpm,torque_nm,tool_wear_min,machine_failure
M,298.1,308.6,1551,42.8,0,0
L,298.2,308.7,1408,46.3,3,0
L,298.1,308.5,1498,49.4,5,0
L,298.2,308.6,1433,39.5,7,0
L,298.2,308.7,1408,40.0,9,0
M,298.1,308.6,1425,41.9,11,0
L,298.1,308.6,1558,42.4,14,0
L,298.1,308.6,1527,40.2,16,0
M,298.3,308.7,1667,28.6,18,0
M,298.5,309.0,1741,28.0,21,0
H,298.4,308.9,1782,23.9,24,0
H,298.6,309.1,1423,44.3,29,0
M,298.6,309.1,1339,51.1,34,0
M,298.6,309.2,1742,30.0,37,0
L,298.6,309.2,2035,19.6,40,0
L,298.6,309.2,1542,48.4,42,0
M,298.6,309.2,1311,46.6,44,0
M,298.7,309.2,1410,45.6,47,0
H,298.8,309.2,1306,54.5,50,0
M,298.9,309.3,1632,32.5,55,0
H,298.9,309.3,1375,42.7,58,0
L,298.8,309.3,1450,44.8,63,0
M,298.9,309.3,1581,30.7,65,0
L,299.0,309.4,1758,25.7,68,0
M,299.0,309.4,1561,37.3,70,0
L,299.0,309.5,1861,23.3,73,0
L,299.1,309.5,1512,39.0,75,0
H,299.1,309.4,1811,24.6,77,0
L,299.1,309.4,1439,44.2,82,0
L,299.0,309.4,1693,30.1,84,0
M,299.1,309.5,1339,48.2,86,0
L,299.0,309.4,1798,25.5,89,0
```

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\*00\_prepare\_data.ipynb x Terminal x 2:ci-iram15-lab1 x 01\_knn\_notebook.ipynb x

Edit in VS Code x Compute: ci-iram15-lab1 - Running

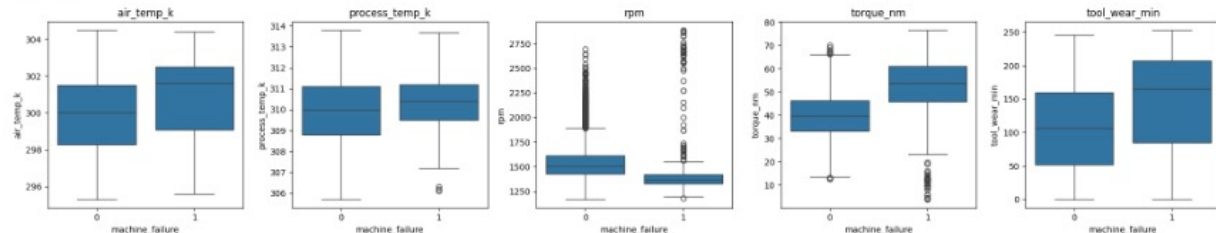
Python 3.10 - SDK v2

ci-iram15-lab1 - Kernel idle Last saved a minute ago Python 3.10 - SDK v2



```
1 fig, axes = plt.subplots(1, 5, figsize=(20, 4))
2 for ax, col in zip(axes, NUM_COLS):
3     sns.boxplot(data=df, x=TARGET, y=col, ax=ax)
4     ax.set_title(col)
5 plt.tight_layout()
6 plt.show()
```

[3] ✓ 1 sec



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Edit in VS Code

Compute: ci-iram15-lab1 - Running

Python 3.10 - SDK v2

ci-iram15-lab1 - Kernel idle

Last saved a few seconds ago

Python 3.10 - SD

4 X\_train, X\_test, y\_train, y\_test = train\_test\_split(5 X, y, test\_size=0.2, stratify=y, random\_state=42)6 print("Train:", X\_train.shape, "failure rate", round(y\_train.mean(), 4))7 print("Test: ", X\_test.shape, "failure rate", round(y\_test.mean(), 4))

[5] ✓ <1 sec

Train: (8000, 6) failure rate 0.0339

Test: (2000, 6) failure rate 0.034

1 def make\_pipeline(scale=True, k=5):2 num\_step = StandardScaler() if scale else "passthrough"3 prep = ColumnTransformer([4 ("num", num\_step, NUM\_COLS),5 ("cat", OneHotEncoder(handle\_unknown="ignore"), CAT\_COLS),6 ])7 return Pipeline([("prep", prep), ("knn", KNeighborsClassifier(n\_neighbors=k))])89 for scale in [False, True]:10 model = make\_pipeline(scale=scale).fit(X\_train, y\_train)11 pred = model.predict(X\_test)12 print(f"Scaling={str(scale):5} accuracy={accuracy\_score(y\_test, pred):.3f} "13 f"recall={recall\_score(y\_test, pred):.3f} F1={f1\_score(y\_test, pred):.3f}")

[6] ✓ <1 sec

Scaling=False accuracy=0.970 recall=0.206 F1=0.318

Scaling=True accuracy=0.974 recall=0.294 F1=0.435

1 |

## Notebooks

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iramshahzadishahzadi892

azureml knn maintenance

&gt; autorml

&gt; data

&gt; deployment

&gt; designer

&gt; notebooks

00\_prepare\_data.ipynb

01\_knn\_notebook.ipynb

&gt; screenshots

.gitignore

ML README.md

requirements.txt

00\_prepare\_data.ipynb x

Terminal x

2:ci iram15 lab1 x

01\_knn\_notebook.ipynb x

Edit in VS Code

Compute:

ci-iram15 lab1 - Running

Python 3.10 - SDK v2

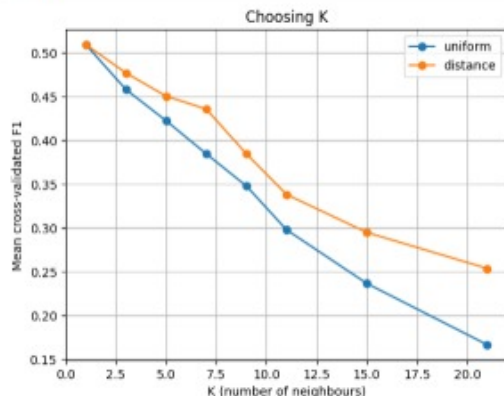
ci-iram15 lab1 - Kernel idle

Last saved a few seconds ago

Best cross-validated F1: 0.580

```
1 res = pd.DataFrame(grid.cv_results_)
2 res = res[res["param_knn_p"] == grid.best_params_["knn_p"]]
3 for w in ["uniform", "distance"]:
4     sub = res[res["param_knn_weights"] == w]
5     plt.plot(sub["param_knn_n_neighbors"].astype(int),
6             sub["mean_test_score"], marker="o", label=w)
7 plt.xlabel("K (number of neighbours)")
8 plt.ylabel("Mean cross-validated F1")
9 plt.title("Choosing K")
10 plt.legend()
11 plt.grid(True)
12 plt.show()
```

[9] ✓ &lt;1 sec





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Default Directory &gt; mlw-lab1-iram15 &gt; Notebooks

## Notebooks

Files Samples



- Users
  - iramshahzadishahzadi892
    - azureml-knn-maintenance
      - automl
      - data
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    - notebooks
      - 00\_prepare\_data.ipynb
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    - requirements.txt

\*00\_prepare\_data.ipynb

Terminal

2-ci-iram15-lab1

01\_knn\_notebook.ipynb



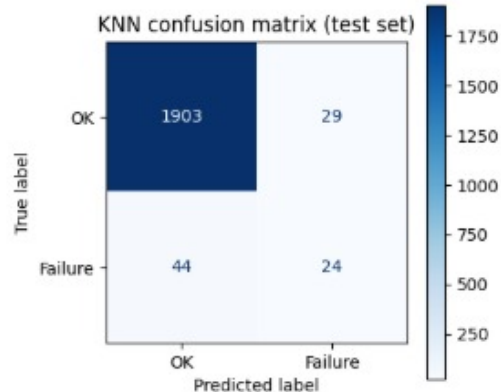
Edit in VS Code

Compute: ci-iram15-lab1 - Running

ci-iram15-lab1 · Kernel idle

Last saved a few seconds ago

```
19 fig.savefig('confusion_matrix.png', bbox_inches='tight')
20 plt.show()
[10] ✓ <1 sec
...
{'accuracy': 0.964, 'precision': 0.453, 'recall': 0.353, 'f1': 0.397, 'auc': 0.669}
precision recall f1-score support
OK 0.98 0.98 0.98 1932
Failure 0.45 0.35 0.40 68
accuracy 0.96 2000
macro avg 0.72 0.67 0.69 2000
weighted avg 0.96 0.96 0.96 2000
```



# knn-notebook-gridsearch Completed

- Overview
- Metrics
- Images
- Child jobs
- Outputs + logs
- Code
- Monitoring

 Refresh  Resubmit  Debug and monitor  Register model  Cancel  Delete  Compare

## Properties

|                                                                                             |                                                                                            |
|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Status                                                                                      | Script name                                                                                |
|  Completed | --                                                                                         |
| Created on                                                                                  | Created by                                                                                 |
| Sep 29, 2026 11:01 PM                                                                       | iram shahzadi                                                                              |
| Start time                                                                                  | Experiment                                                                                 |
| Sep 29, 2026 11:01 PM                                                                       | <a href="#">knn-predictive-maintenance</a>                                                 |
| Duration                                                                                    | Arguments                                                                                  |
| 18.20s                                                                                      | None                                                                                       |
| Compute duration                                                                            | Registered models                                                                          |
| 18.20s                                                                                      | <a href="#">knn-maintenance-notebook:1</a>                                                 |
| Compute target                                                                              | See all properties                                                                         |
| None                                                                                        |  Raw JSON |
| Name                                                                                        |                                                                                            |
| 9753c06b-fe8f-4fe9-a392-1fa85715c61d                                                        |                                                                                            |

## Outputs

Output name: [mlflow\\_log\\_model\\_1939882990](#)  
Model: [azureml\\_9753c06b-fe8f-4fe9-a392-1fa85715c61d\\_output\\_mlflow\\_log\\_model\\_1939882990:1](#)  
Asset URI:

## Tags

[mlflow.runName : knn-notebook-gridsearch](#) [mlflow.user : azureuser](#)

## Params

[knn\\_n\\_neighbors : 1](#) [knn\\_p : 1](#) [knn\\_weights : uniform](#) [scaler : StandardScaler](#)

## Metrics

|               |                |
|---------------|----------------|
| cv_best_f1    | test_f1        |
| 0.5090390     | 0.3966942      |
| test_accuracy | test_precision |
| 0.9635        | 0.4528302      |
| test_auc      | test_recall    |
| 0.6689654     | 0.3529412      |

- Manage
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## Model List

[+ Register](#)
[Refresh](#)
[Delete](#)
[Archive](#)
[Use this model](#)
[Compare](#)
[Reset view](#)

☒ Show latest versions only ☒ Include archived

 Search

Filter Columns

| Name                     | ☆ | Version | Type   | Source ▾       | Experiment                 | Job (Run ID)                    | Created on ↓          | Tags ▾ | Properties ▾ |
|--------------------------|---|---------|--------|----------------|----------------------------|---------------------------------|-----------------------|--------|--------------|
| knn-maintenance-notebook |   | 1       | MLFLOW | This workspace | knn-predictive-maintenance | 9753c06b-fe8f-4fe9-a392-1fa8... | Sep 29, 2026 11:02 PM |        | ...          |

### Submit an Automated ML job

- ✔ Training method

## 2 Basic settings

### 3 Task type & data

#### 4 Task settings

5 Compute

## 6 Review

### Basic settings

Let's start with some basic information about your training job.

Job name \*

knn-automi-iram15

Experiment name \*

☐ Select existing ☒ Create new

New experiment name \*

knn-predictive-maintenance-automl

### Description

## Tags

| Name |
|------|
|------|

|   |       |
|---|-------|
| : | Value |
|---|-------|

Add

Default Directory > mlw-lab1-iram15 > Training job

Submit an Automated ML job

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Training method

Basic settings

Task type & data

Task settings

Compute

Review

Task type & data

Choose the type of task that you would like your model to perform and the data to use for training. [Learn more](#)

Select task type \*  
Classification

Select data

Make sure your data is preprocessed into a supported format.

+ Create

Refresh

Show supported data assets only

Reset view

Search

Filter

Columns

| Name                   | Type  | Created on           | Modified on          |
|------------------------|-------|----------------------|----------------------|
| ai4i-maintenance-table | Table | Sep 29, 2026 3:32 PM | Sep 29, 2026 3:32 PM |

<< < Page 1 of 1 > >>

25/Page

## Submit an Automated ML job

- ✓ Training method
- ✓ Basic settings
- ✓ Task type & data
- 4 Task settings**
- 5 Compute
- 6 Review

### Task settings

Task type

Classification

Data

ai4i-maintenance-table [\(View data\)](#)

Target column \*

machine\_failure (String)

Classification settings

☐ Enable deep learning ⓘ

[View additional configuration settings](#) [View featurization settings](#)

[Limits](#)

### Validate and test

You can choose a validation type and select test data as an optional step.

Validation type ⓘ

Automatic

Test data ⓘ

None

[Back](#)

[Next](#)

## Additional configuration

Primary metric ⓘ

AUCWeighted

☐ Explain best model ⓘ

☐ Enable ensemble stacking ⓘ

☐ Use all supported models

Allowed models ⓘ

KNN

Positive class label ⓘ

Positive class label

[Save](#)

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## Submit an Automated ML job

[✓ Training method](#)[✓ Basic settings](#)[✓ Task type & data](#)[4 Task settings](#)[5 Compute](#)[6 Review](#)

### Metric score threshold ⓘ

### Experiment timeout (minutes) ⓘ

### Iteration timeout (minutes) ⓘ

☒ [Enable early termination ⓘ](#)

### Validate and test

You can choose a validation type and select test data as an optional step.

### Validation type ⓘ

### Number of cross validations \* ⓘ

### Test data ⓘ

### Percentage test of data \* ⓘ

Automated ML recommends that between 10 and 30 percent of data is held out for test

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Default Directory > mlw-lab1-iram15 > Jobs > knn-automl-fixed > hungry\_ear\_sk80gs8hk7

hungry\_ear\_sk80gs8hk7    Completed

Overview

Data guardrails

Models + child jobs

Outputs + logs

Child jobs

Refresh

Edit and submit (preview)

Register model

Cancel

Delete

Compare

## Properties

Status

 Completed

Created on

Sep 30, 2026 7:50 AM

Start time

Sep 30, 2026 7:50 AM

Duration

15m 11.49s

Compute duration

15m 11.49s

Name

hungry\_ear\_sk80gs8hk7

Script name

--

Created by

iram shahzadi

Job type

Automated ML

Experiment

knn-automl-fixed

Arguments

None

See all properties

 Raw JSON

See YAML job definition

 Job YAML

## Tags

\_azureml.ComputeTargetType

fit\_time\_000 :

0.07290619999999999;0.516832;0.045303600000000000;0.016553999999999996;0.0807524;0.46207160000000001;0.07118580000000001;0.087906400000000001;0.1492934;6

iteration\_000 : 0;1;2;3;4;5;6;7;8;9

## Inputs

Input name: training\_data

Data asset: ai4i-maintenance-fixed:1

Asset URI: azureml:ai4i-maintenance-fixed:1

## Outputs

Output name: best\_model

Model: azureml\_hungry\_ear\_sk80gs8hk7\_9\_output\_mflow\_log\_model\_1137969628:1

Asset URI: azureml:azureml\_hungry\_ear\_sk80gs8hk7\_9\_output\_mflow\_log\_model\_1137969628:1

## Best model summary

Algorithm name

VotingEnsemble

Ensemble details

 View ensemble details

AUC weighted

0.97330  View all other metrics

Sampling

100.00 % 

Registered models

No registration yet

Deploy status

No deployment yet

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hungry\_ear\_sk80gs8hk7    Completed

Overview Data guardrails Models + child jobs Outputs + logs Child jobs

 Refresh  Edit and submit (preview)  Register model  Cancel  Delete

Data guardrails are run by Automated ML when automatic featurization is enabled. This is a sequence of checks over the input data to ensure high quality data is being used to train model.

| Type                                                                                                      | Status  | Description                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Class balancing detection                                                                                 | Alerted | Imbalanced classes were detected in your inputs.<br><a href="#">Learn more about imbalanced data.</a>                                              |
|  View additional details |         |                                                                                                                                                    |
| Missing feature values imputation                                                                         | Passed  | No feature missing values were detected in the training data.<br><a href="#">Learn more about missing value imputation.</a>                        |
| High cardinality feature detection                                                                        | Passed  | Your inputs were analyzed, and no high cardinality features were detected.<br><a href="#">Learn more about high cardinality feature detection.</a> |

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hungry\_ear\_sk80gs8hk7

Completed

Overview

Data guardrails

Models + child jobs

Outputs + logs

Child jobs

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Explain model

View generated code

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| Algorithm name                           | Responsible AI | AUC weighted | Sampling | Created on           | Duration | Hyperparameter                      |
|------------------------------------------|----------------|--------------|----------|----------------------|----------|-------------------------------------|
| VotingEnsemble                           |                | 0.97330      | 100.00 % | Sep 30, 2026 8:04 AM | 50s      | algorithm : ['LightGBM', 'XGBoost'] |
| MaxAbsScaler, LightGBM                   |                | 0.97125      | 100.00 % | Sep 30, 2026 7:57 AM | 33s      | min_data_in_leaf : 20               |
| MaxAbsScaler, XGBoostClassifier          |                | 0.97006      | 100.00 % | Sep 30, 2026 7:57 AM | 33s      | tree_method : auto                  |
| MaxAbsScaler, RandomForest               |                | 0.95542      | 100.00 % | Sep 30, 2026 7:57 AM | 30s      | bootstrap : true class_weight :     |
| StandardScalerWrapper, LightGBM          |                | 0.94662      | 100.00 % | Sep 30, 2026 7:57 AM | 36s      | boosting_type : gbdt colsample :    |
| SparseNormalizer, XGBoostClassifier      |                | 0.93510      | 100.00 % | Sep 30, 2026 7:57 AM | 30s      | booster : gbtree colsample_by :     |
| SparseNormalizer, XGBoostClassifier      |                | 0.93398      | 100.00 % | Sep 30, 2026 7:57 AM | 33s      | booster : gbtree colsample_by :     |
| StandardScalerWrapper, XGBoostClassifier |                | 0.92909      | 100.00 % | Sep 30, 2026 7:57 AM | 31s      | booster : gbtree colsample_by :     |
| SparseNormalizer, RandomForest           |                | 0.92324      | 100.00 % | Sep 30, 2026 7:57 AM | 39s      | bootstrap : true class_weight :     |
| MaxAbsScaler, RandomForest               |                | 0.89734      | 100.00 % | Sep 30, 2026 7:57 AM | 37s      | bootstrap : true class_weight :     |

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Default Directory > mlw-lab1-iram15 > Designer > Authoring

Execute Python Script

Tags: All Add filter

DataComponent

1 + Most relevant

Execute Python Script

Microsoft

Execute a Python script from an Azure Machine Learning designer pipeline. (Learn More)(https://ak...

azureml.Designer>true

3/20/2025

Undo

Redo

Validate

Show lineage

Clone

AutoSave

Configure & Submit

knn-designer-iram

Save

Pipeline interface

ai4i-maintenance-fixed

ai4i-maintenance-fixed

1

Data output

Dataset

Split Data

split\_data

Results datas...

Results datas...

Dataset1

Dataset2

Script b...

Execute Python Script

execute\_python\_script

Parameters

Result datase...

Result datase...

Navigator

100%

...

Execute Python Script

6 import pandas as pd

7

8 # The entry point function MUST have two input arguments.

9 # If the input port is not connected, the corresponding

10 # dataframe argument will be None.

11 # Param<dataframe1>: a pandas.DataFrame

12 # Param<dataframe2>: a pandas.DataFrame

13 def azureml\_main(dataframe1 = None, dataframe2 = None):

14

15 # Execution logic goes here

16 print(f'Input pandas.DataFrame #1: {dataframe1}')

17

18 # If a zip file is connected to the third input port,

19 # it is unzipped under ".\Script Bundle". This directory is ad

20 # to sys.path. Therefore, if your zip file contains a Python f

21 # mymodule.py you can import it using:

22 # import mymodule

23

24 # Return value must be of a sequence of pandas.DataFrame

25 # E.g.

Edit code

Output settings

Input settings

Run settings

Node information

Component information

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Outline

Type node name, comment or comp...

Add filter

knn-designer-iram

ai4i-maintenance-fixed

split\_data

execute\_python\_script

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Publish

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Show lineage

Show compare list

Create inference pipeline

Delete

Cancel

knn-designer-iram

Completed

Share

Add to compare

Job overview

ai4i-maintenance-fixed

1

Data output

Dataset

Split Data

split data

Results datas...

Results datas...

Dataset1

Dataset2

Script b...

Execute Python Script

execute python script

Result datase...

Result datase...

Navigator

100%





ci-iram15-lab1.eastus.instances.azureml.ms/lab

File Edit View Run Kernel Tabs Settings Help

azureml-knn-maintenance / notebooks /

| Name                     | Modified  |
|--------------------------|-----------|
| 00_prepare_data.ipynb    | 4h ago    |
| 00_prepare_data.ipynb... | 4h ago    |
| 01_knn_notebook.ipynb    | yesterday |
| 01_knn_notebook.ipynb... | yesterday |
| 02_deploy_endpoint.i...  | 11m ago   |
| 02_deploy_endpoint.i...  | 2h ago    |
| conda.yaml               | 41m ago   |
| confusion_matrix.png     | yesterday |
| score.py                 | 20m ago   |

02\_deploy\_endpoint.ipynb

Python 3.10 - SDK v2

```
        "columns": [
            "air_temp_k",
            "process_temp_k",
            "rpm",
            "torque_nm",
            "tool_wear_min",
            "type",
        ],
        "data": [
            [298.1, 308.6, 1551.0, 42.8, 5.0, "M"]
        ],
    }
}

body = str.encode(json.dumps(input_data))
headers = {
    "Content-Type": "application/json",
    "Authorization": f"Bearer {primary_key}",
}

req = urllib.request.Request(scoring_uri, body, headers)

try:
    response = urllib.request.urlopen(req)
    result = response.read().decode("utf-8")
    print("Prediction Response:", result)
except urllib.error.HTTPError as error:
    print(f"Request failed with status code: {error.code}")
    print(error.read().decode("utf-8"))

Prediction Response: {"prediction": [0]}
```

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Default Directory > mlw-lab1-iram15 > Endpoints > knn-maint-iram15-v3

## knn-maint-iram15-v3

Details Test Consume Monitoring PREVIEW Logs

Deployment

blue

Review the original model card to understand the inputs, outputs, data used to train the model, evaluation metrics, license, intended uses, limitations, and bias before using the model.

[View model](#)

### Sample inference

Input \*

{ }

Test



Search





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Default Directory > mlw-lab1-iram15 > Endpoints > knn-maint-iram15-v3

## knn-maint-iram15-v3

Details Test Consume Monitoring PREVIEW Logs

### Basic consumption info

#### REST endpoint

<https://knn-maint-iram15-v3.eastus.inference.ml.azure.com/score>

#### Authentication

Primary key

.....



[Regenerate](#)

Secondary key

.....



[Regenerate](#)

### Consumption option

#### Consumption types

JavaScript

Python

C#

JSON

```
1 // Request data goes here
2 // The example below assumes JSON formatting which may be updated
3 // depending on the format your endpoint expects.
4 // More information can be found here:
5 // https://docs.microsoft.com/azure/machine-learning/how-to-deploy-advanced-entry-script
6 const requestBody = `{}`;
7
8 const requestHeaders = new Headers({"Content-Type" : "application/json"});
9
10 // Replace this with the primary/secondary key, AMLToken, or Microsoft Entra ID token for the endpoint
11 const apiKey = "";
12 if (!apiKey)
13 }
```

Microsoft Azure

Upgrade

Search resources, services, and docs (G+)

Copilot

iramshahzadishahzadi8...  
DEFAULT DIRECTORY (IRAMSHA...

Home > Resource Manager | Resource groups

Resource Manager | Resource groups

Default Directory

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Name ↑

rg-mlab1-iram15

Showing 1 - 1 of 1. Display count:

au...

rg-mlab1-iram15

Resource group

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Analyze (Preview)

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Delete a resource group

The following resource group and all its dependent resources will be permanently deleted.

Resource group to be deleted

rg-mlab1-iram15

12 resources being deleted across 1 subscriptions, 1 resource groups

All dependent resources, including hidden types, are shown

| Name                                           | Resource type                      | Subscription         | Resource Group  |
|------------------------------------------------|------------------------------------|----------------------|-----------------|
| Application Insights Smart Detection           | Action group                       | Azure subscription 1 | rg-mlab1-iram15 |
| blue (mlw-lab1-iram15/knn-maint-iram15-v3/blue | Machine learning online deployment | Azure subscription 1 | rg-mlab1-iram15 |
| e924bdabd40e4e8198267af930adb08c               | Container registry                 | Azure subscription 1 | rg-mlab1-iram15 |
| knn-maint-iram15 (mlw-lab1-iram15/knn-maint-i  | Machine learning online endpoint   | Azure subscription 1 | rg-mlab1-iram15 |
| knn-maint-iram15-v2 (mlw-lab1-iram15/knn-mair  | Machine learning online endpoint   | Azure subscription 1 | rg-mlab1-iram15 |
| knn-maint-iram15-v3 (mlw-lab1-iram15/knn-mair  | Machine learning online endpoint   | Azure subscription 1 | rg-mlab1-iram15 |
| knn-maint-iram15b (mlw-lab1-iram15/knn-maint-  | Machine learning online endpoint   | Azure subscription 1 | rg-mlab1-iram15 |
| mlw-lab1-iram15                                | Azure Machine Learning workspace   | Azure subscription 1 | rg-mlab1-iram15 |
| mlwlab1iram152381727715                        | Log Analytics workspace            | Azure subscription 1 | rg-mlab1-iram15 |

Enter resource group name to confirm deletion \*